

Getting Started with Git and GitHub pt. 1

Coffee, Cookie and Coding (C3)
Workshop supported by the Public
Health Data Science and Data
Equity team

Shelby Golden, M.S.

September 29th, 2025



Yale SCHOOL OF PUBLIC HEALTH
Data Science and Data Equity



[xkcd Comic 1597 - Git](#) Downloaded April 26th, 2025.



Shelby Golden, M.S.

- Worked 7 years as a Molecular Biologist and Biochemist.
- Received a Masters in Applied Computational Mathematics from Johns Hopkins University in 2024.





Justin DeMayo

- System and Application Specialist for the Harvey Cushing/John Hay Whitney Medical Library.
- Offers introductory command line, git, and python courses.



Today's Learning Objectives

- 01 Understand the purpose and value of Git and GitHub in managing coding projects. (~ 5 minutes)
- 02 Learn how Git manages files for version control locally and distributes them through GitHub. (~ 15 minutes)
- 03 Set up and configure your local Git and GitHub accounts using either HTTPS or SSH Keys. (~ 40 minutes)

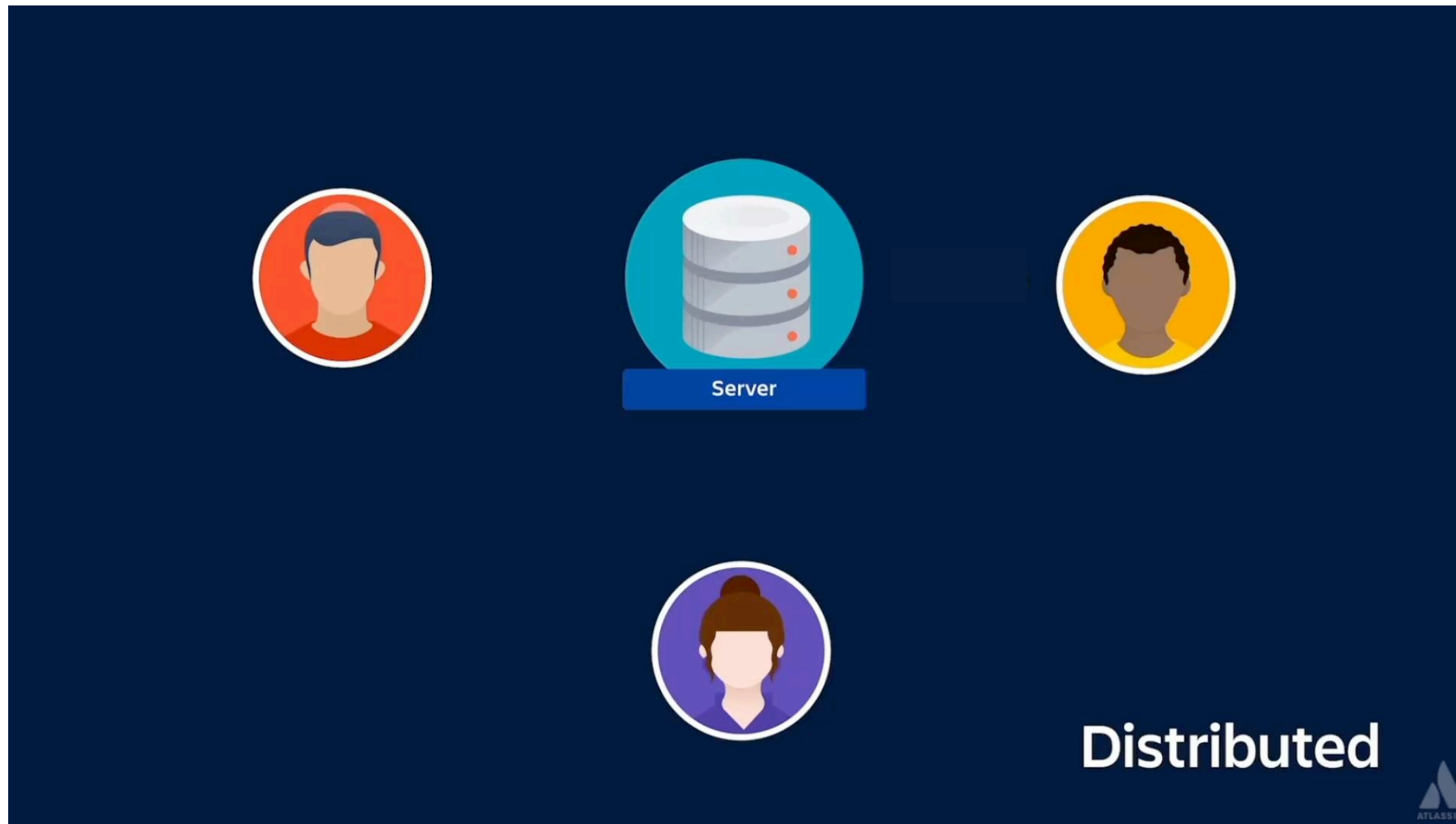


Our Choice Resources

- Yale's Center for Research Computing workshop [Version Control by Git](#) by [Kaylea Nelson](#)
- Yale's Harvey Cushing/John Hay Whitney Medical Library workshop [Git & GitHub: An Introduction To Version Control](#) by [Justin DeMayo](#)
- [Getting Git Right](#) by [Atlassian](#)
- [Git and GitHub Tutorial](#) by [W3 Schools](#)
- [Introduction to GitHub](#) by [GitHub](#)
- [Happy Git and GitHub for useR](#) by [Jenny Bryan](#)

What is Git and GitHub?





[What is Version Control](#) by Atlassian. Updated February 23rd, 2020.

[Git logo](#). Downloaded October 10th, 2024.

[GitHub logo](#). Downloaded October 10th, 2024.



System for project management by distributive version control (DVCS).

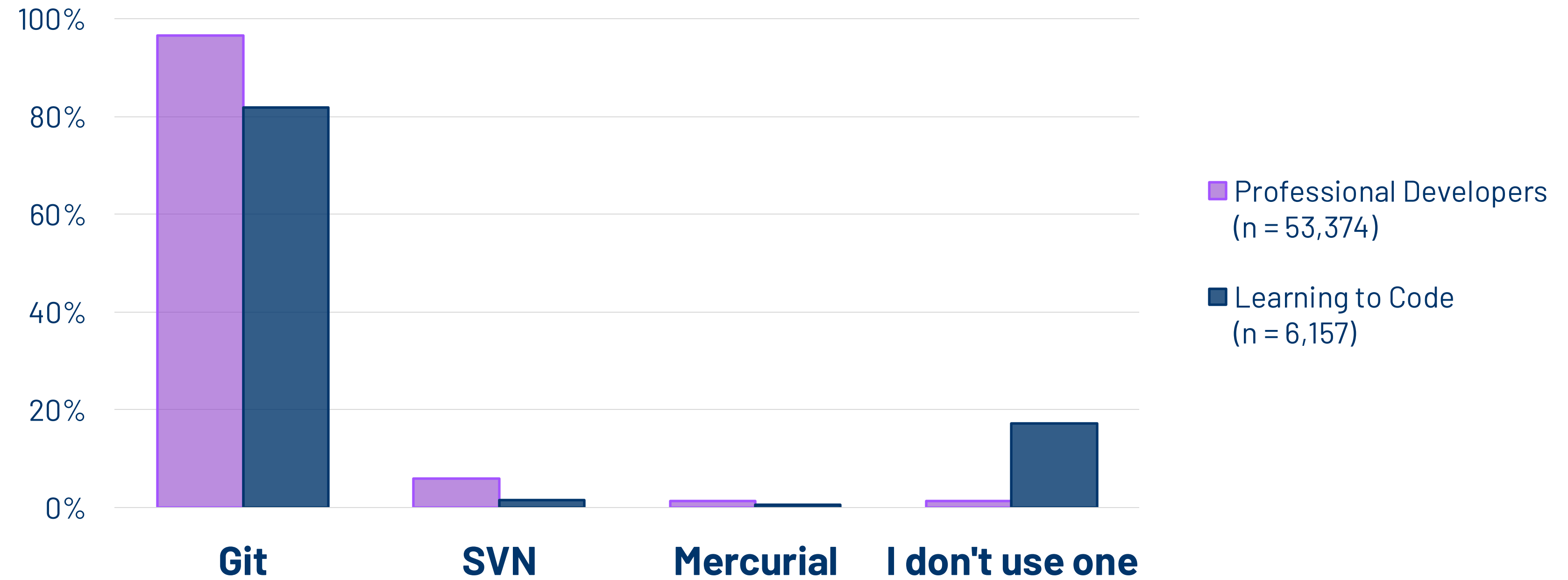


Developer platform for housing and managing projects and acts as the DVCS server.

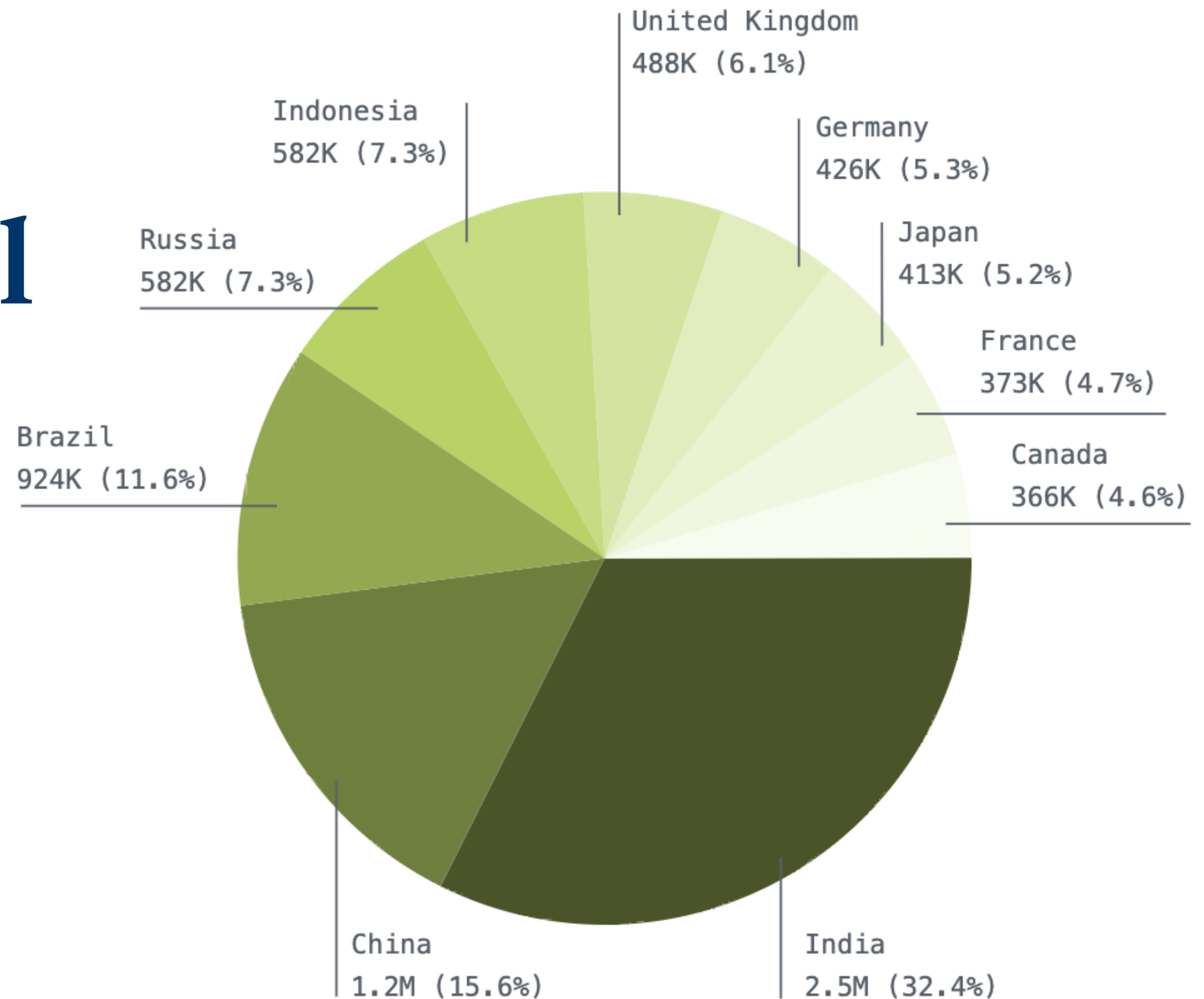


Git is the most widely used VCS but is underutilized by those learning to code.

[2022 Developer Survey](#) by StackOverflow. Published 2022.



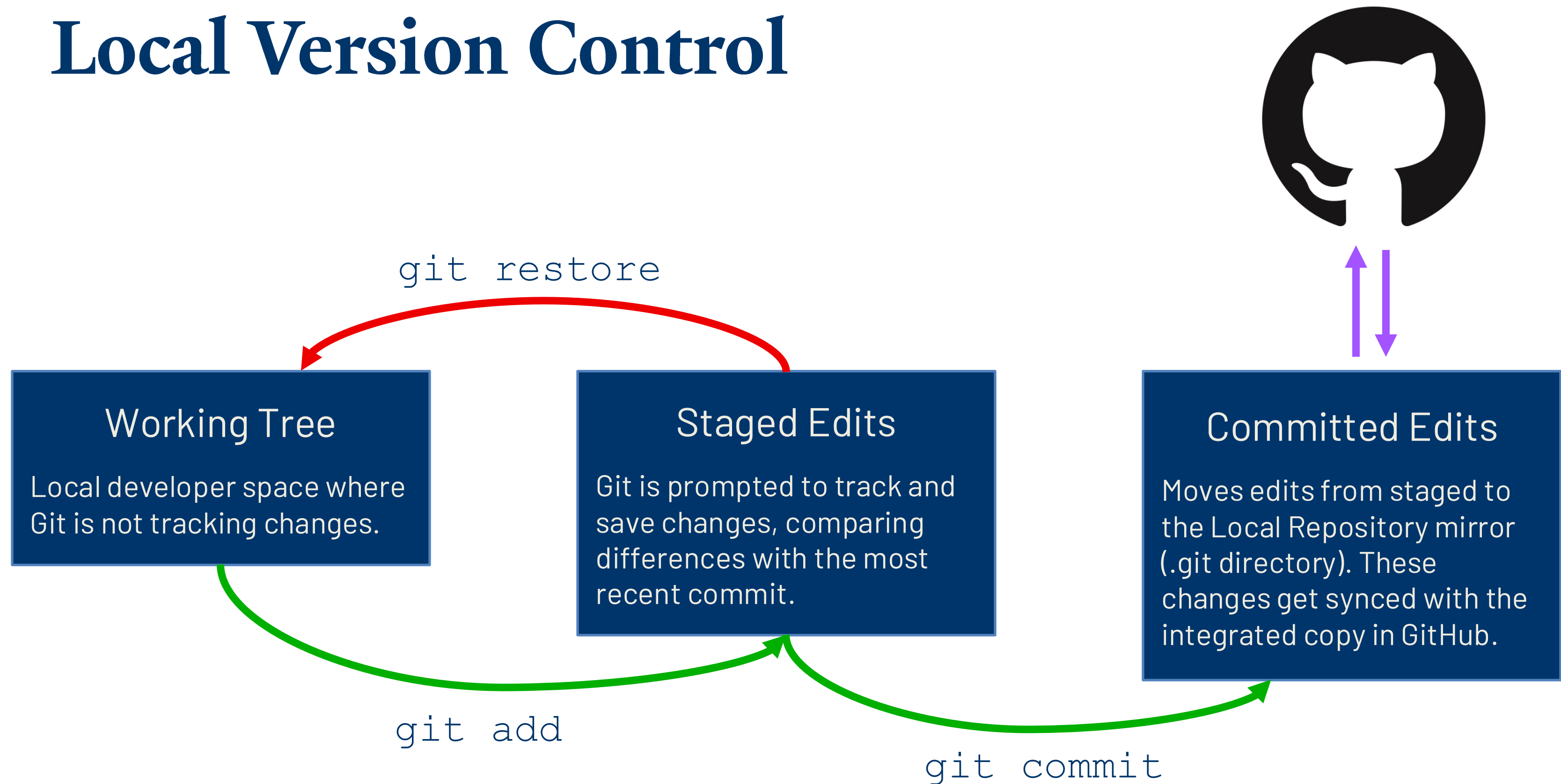
GitHub is an invaluable tool for coders worldwide.



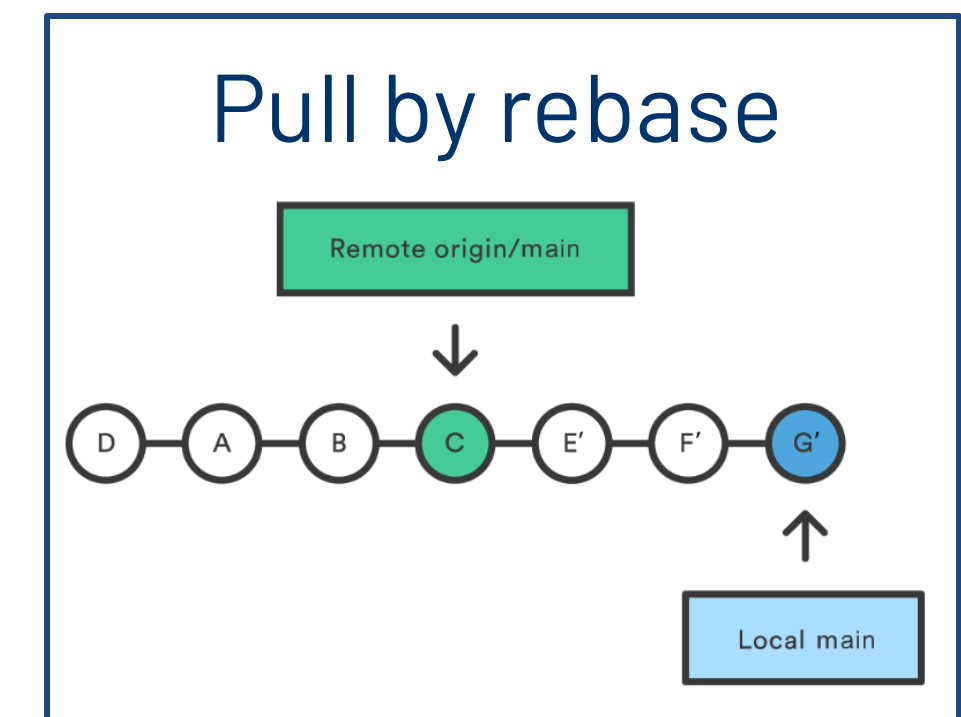
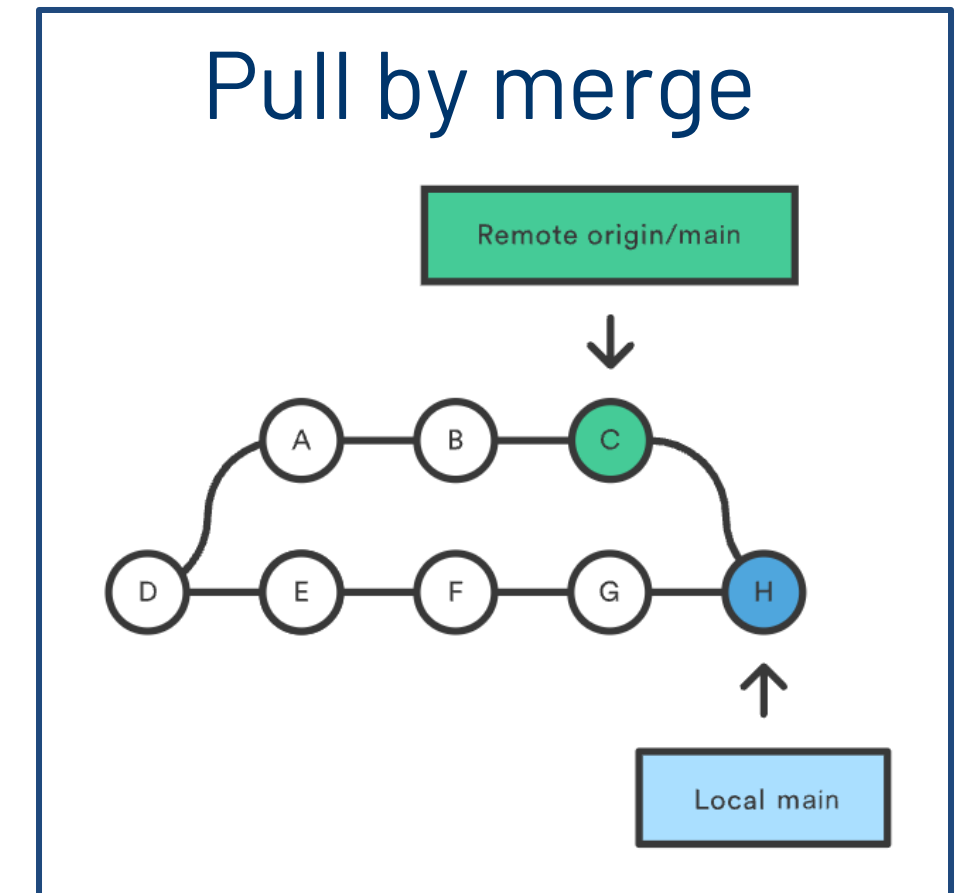
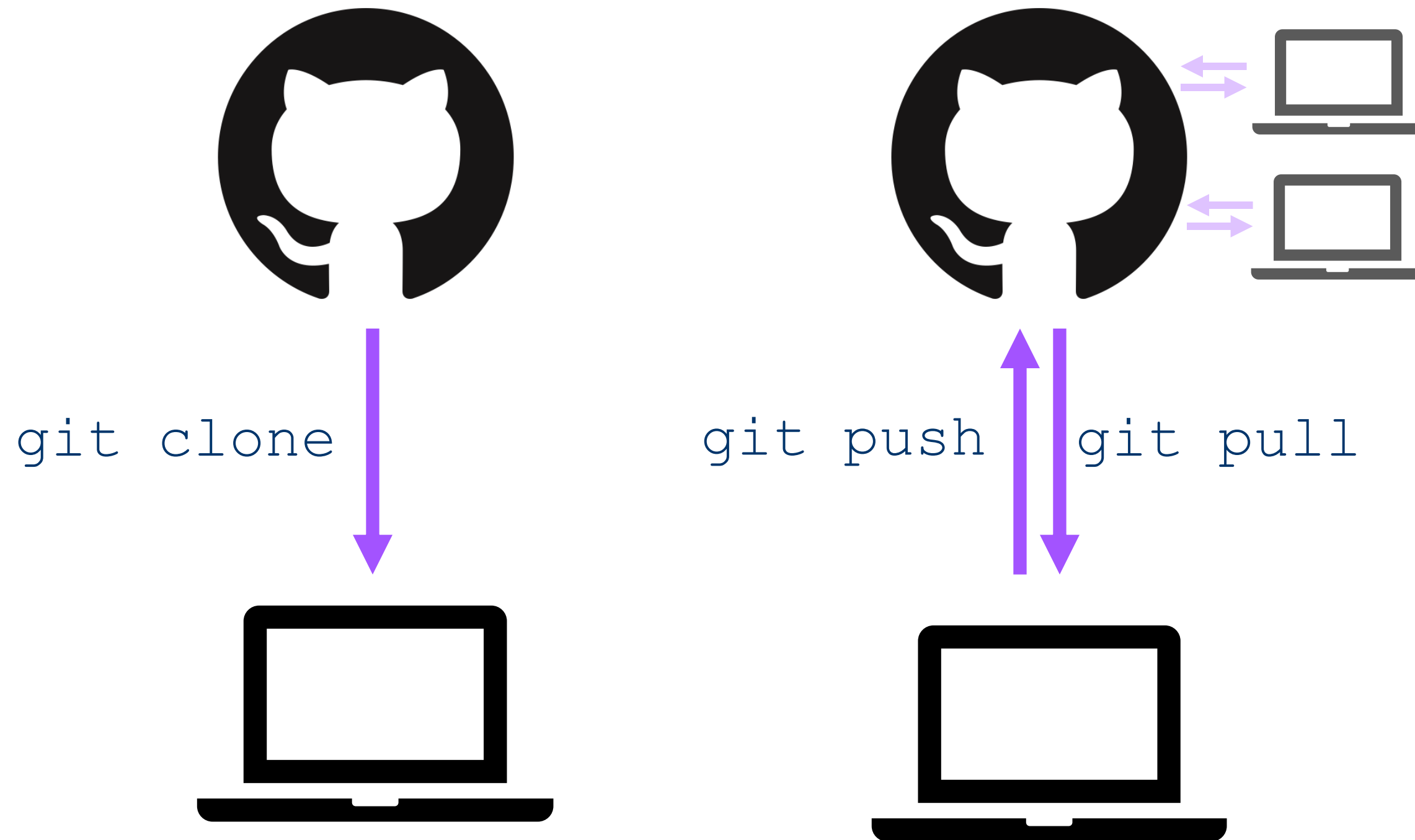
How do Git and GitHub relate to one another?



Local Version Control



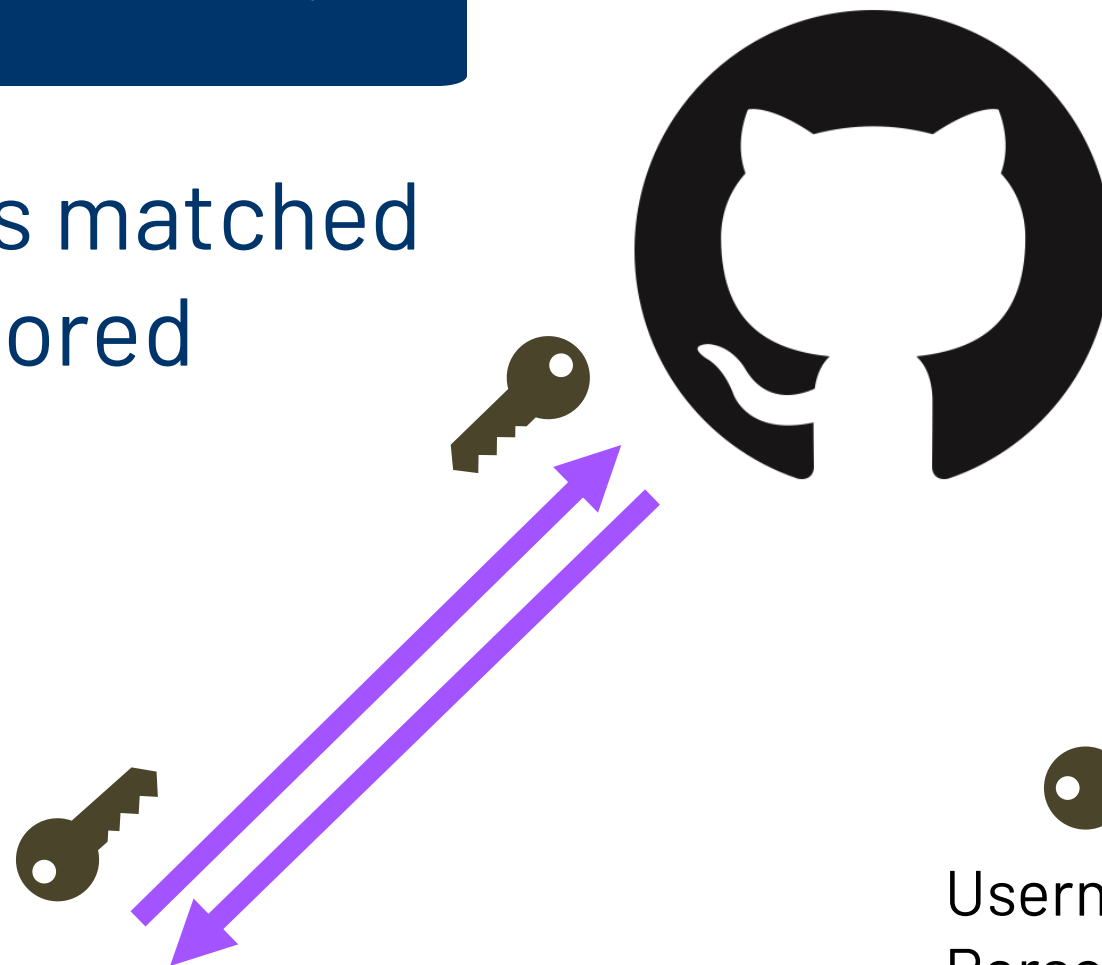
Remote Storage and Distribution



Secure Shell (SSH) Keys

A local **private key** is matched with the remotely stored **public key**.

Keys can decrypt messages sent by their pair.

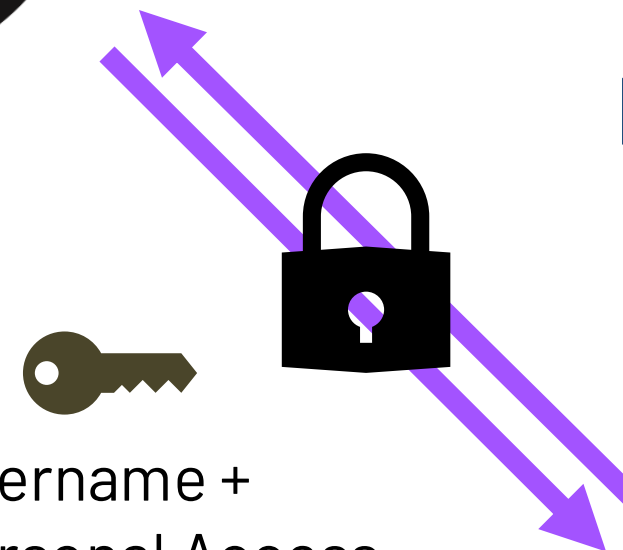


Hypertext Transfer Protocol Secure (HTTPS)

Information is securely transferred over the internet.

Log-in credentials verify access permissions to the repository.

Username +
Personal Access
Token (PAT)



Additional Helpful Articles

- [Cloning with SSH URL's](#) and [Troubleshooting SSH](#) by GitHub
- [Cloning with HTTPS URL's](#) and [Managing your PATs](#) by GitHub
- [Commit signature verification](#) (sign [commits](#) and [tags](#)) by GitHub
- [Managing deploy keys](#) by GitHub
- [Switching remote URLs from SSH to HTTPS](#) and [SSH to HTTPS](#) by GitHub
- [Using SSH agent forwarding](#) by GitHub
- [Caching your GitHub credentials in Git](#) for Windows by GitHub and [Install the git-credential-osxkeychain helper](#) for Macs by Atlassian

Credential Configuration: Let's Get Started!



Essential Setup



- Install Git
- GitHub Account
- HTTPS Personal Access Token (PAT)
- SSH Key Pairs



Data Transfers



- Install Git
- GitHub Account
- HTTPS Personal Access Token (PAT)
- SSH Key Pairs



Getting Started with Git and GitHub - DSDE Event Evaluation



Public Health Data Science & Data Equity 2025 Calendar

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WORKSHOPS / TUTORIALS

- 09.09.25 • 11 AM - 1 PM
AI Agents for Biomedical Discoveries Tutorial
w/ James Zou, PhD, Stanford University • Kline Tower Rm 1327 (219 Prospect St)
- 09.29.25 • 3 PM - 4 PM
Getting Started with Git & GitHub Part 1 Workshop
w/ Shelby Golden, MS • LEPH Rm. 103 (60 College Street)
- 11.21.25 • 9:30 AM - 11 AM
Let's talk Data: How to use the NIH All of Us Tutorial
w/ Maxwell Salvatore, PhD, University of Pennsylvania • Winslow Auditorium (60 College Street)
- 11.24.25 • 3 PM - 4 PM
Data Visualization with ggplot2 Workshop
w/ Shelby Golden, MS • LEPH Rm. 101 (60 College Street)

LECTURES

- 10.24.25 • 12:30 PM - 2 PM
The Joy of Stats (the Missing Data) Journey Lecture
w/ Rod Little, PhD, University of Michigan • Winslow Auditorium (60 College Street)
- 11.07.25 • 1 PM - 2 PM
A Weird, Wild Ride Journey Lecture
w/ Gregg Gonsalves, PhD, Yale School of Public Health • Winslow Auditorium (60 College Street)

COURSES

- Fall 2025 / Monday 5 PM - 6 PM
BIS 564: Precision Medicine Modeling: A Primer
w/ Brian Tom, PhD, University of Cambridge • George Saden Visiting Faculty
- Spring 2026 / Wednesday 5 PM - 6 PM
BIS 563: Causal Reasoning & Inference for Health & Social Scientists
w/ Peter Tennant, PhD, University of Leeds • George Saden Visiting Faculty
- Spring 2026 / Monday 1 PM - 2:50 PM
BIS 565: The Role of Ethics and Equity in Data Science and AI
w/ Bhramar Mukherjee, PhD, Yale School of Public Health • Senior Associate Dean for Public Health Data Science and Data Equity, Anna MR Lauder Professor of Biostatistics



For an updated list of events,
including registration details,
please visit our website:
sph.yale.edu/dsde

Thank you!

ysph.yale.edu
sph.yale.edu/dsde

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Public Health Data Science and Data Equity
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Appendix

Glossary

Version Control Manage, organize, and track different versions of files. Identify differences between versions and allows reverting to older versions.

Distributive Version Control System The project codebase is copied as a mirror to each contributor's local computer. Local changes get synched via patches sent peer-to-peer through the server.

Command-Line Interface (CLI) A texted-base application that directly interacts with the computer's operating system, manages files, and can run programs. It typically lacks a graphical user interface (GUI).

Glossary

Shell A program used by the CLI to mediate communication between the user and computer by interpreting commands and outputs. Examples: Bash, PowerShell, etc.

Mirror An exact copy of a project from a server, including a full-change history.

Server Computer or system that provides resources (i.e. data or programs) to other computers, known as clients, over a network.

Patch Snippets of code or data used to update existing software.

Glossary

Peer-to-peer Participants in a network act as both client and server by trading resources and services with one another.

git add Prompt git to track changes that have been made to specific files and compare those differences to previously saved version in the .git directory.

git commit Save your changes as a snapshot in your project's history, present in the .git directory.

git push Upload your committed work to a remote repository, so others can see it.

Glossary

git pull Download changes from a remote repository and merge them into your local branch.

git fetch Retrieve changes from a remote repository without merging them into your local branch.

git merge One method to reconcile different committing histories in divergent branches. Creates a new version integrating the head of the two branches in a three-way commit.

git rebase An alternative to merge. The branch commit histories are realigned so that the leading one defines the commit parent history of the following branch, thus rebasing its commits.

Glossary

git clone Make a copy of a project from the internet to your own computer.

git branch Create or list different “versions” or “paths” of a project that you can work on separately.

git checkout Command to switch between different the branches.

Glossary

Helpful Cheat Sheets:

1. [Git Cheat Sheet](#) by Atlassian
2. [Vim Cheat Sheet](#) by Richard Torruellas
3. [Bash Shortcuts](#) by Mohan Balasundaram
4. [Command Line Cheat Sheet](#) by Tobias Günther

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2. J. Bryan and J. Hester, "Let's Git started," Happy Git and GitHub for the useR. Accessed: Oct. 09, 2024. [Online]. Available: <https://happygitwithr.com/>
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